Homework 01

1. A stock's price at time t is $P_t = \$100$. Over the next three periods, the continuously compounded returns are:

$$r_{t+1} = 0.1$$
$$r_{t+2} = -0.05$$
$$r_{t+3} = 0.02$$

Calculate stock's price at times $t+1, t+2, t+3$

$$r_{t+1} = \ln\left(\frac{P_{t+1}}{P_t}\right) = \ln(P_{t+1}) - \ln(P_t)$$

$$0.1 = \ln\left(\frac{P_{t+1}}{100}\right) \Rightarrow e^{0.1} = \frac{P_{t+1}}{100}$$

$$P_{t+1} = e^{0.1} \cdot 100 = 1.1052 \times 100 = 110.52\$$$

$$-0.05 = \ln\left(\frac{P_{t+2}}{P_{t+1}}\right) = \ln\left(\frac{P_{t+2}}{110.52}\right)$$

$$\Rightarrow e^{-0.05} = \frac{P_{t+2}}{110.52} \Rightarrow P_{t+2} = e^{-0.05} \times 110.52 =$$
$$= 0.9512 \times 110.52$$
$$= 105.13\$$$

$$0.02 = \ln\left(\frac{P_{t+3}}{P_{t+2}}\right) = \ln\left(\frac{P_{t+3}}{105.13}\right)$$

$$\Rightarrow e^{0.02} = \frac{P_{t+3}}{105.13} \Rightarrow P_{t+3} = e^{0.02} \times 105.13 =$$
$$= 1.020 \times 105.3$$
$$= 107.4\$$$

2. Assume the simple returns over the same periods are $R_{t+1} = 0.1$, $R_{t+2} = -0.05$ and $R_{t+3} = 0.02$

Calculate the stock price at times $t+1, t+2, t+3$

$$R_{t+1} = \frac{P_{t+1}}{P_t} - 1$$

$$0.1 = \frac{P_{t+1}}{100} - 1 \Rightarrow P_{t+1} = 1.1 \times 100 = 110\$$$

$$-0.05 = \frac{P_{t+2}}{P_{t+1}} - 1 \Rightarrow P_{t+2} = 0.95 \times 110 = 104.5\$$$

$$0.02 = \frac{P_{t+3}}{P_{t+2}} - 1 \Rightarrow P_{t+3} = 1.02 \times 104.5 = 106.59\$$$

3. Compare the final prices obtained using continuously compounded returns and simple returns.

Text

a) Are they the same?

No, they are not exactly the same.

$r_{t+3} > R_{t+3} \Rightarrow$ the continuously compounded approach